Name: 4.3 Animal or environmental wellbeing

<Files\\Processed x Highly relevant\\2024-06-21 - HBT - AI boost for farm productivity> - § 1 reference coded [2.09% Coverage]

Reference 1 - 2.09% Coverage

Global farming technology company GEA collaborated with local start-up AgriAI to launch a new system called the walkover teat sprayer at Fieldays last week to combat the problem of mastitis in dairy herds.

<Files\\Processed x Highly relevant\\2024-07-24 - NZH - Silver Fern Farms> - § 2 references coded [9.44% Coverage]

Reference 1 - 3.09% Coverage

This rewards farmers who actively care for the land through environmental practices.

Reference 2 - 6.36% Coverage

Certification is also reliant on Silver Fern Farms continued effort to reduce Scope 1 & 2 emissions within its operational boundary. In 2023 these emissions reduced by 6.7%.

<Files\\Processed x Highly relevant\\2024-07-31 - NAD - Future farming- Why not combin> - § 3 references coded [21.33% Coverage]

Reference 1 - 4.03% Coverage

New technology using lasers and artificial intelligence software is helping growers get on top of their weeding, without the need for environmentally harmful chemicals.
Weed c

Reference 2 - 5.90% Coverage

â€œIts goal is to offer farmers a precise, efficient and environmentally friendly solution to weed management and enhancing crop yields and reducing their reliance on chemical herbicides.â€ The AI software allows it to be â€œvery targetedâ€ , he said. â€œHow

Reference 3 - 11.39% Coverage

â€œIt even identifies weeds that you want to kill from weeds you donâ€™t want to kill and then that AI guides the laser to the exact position of the weed and that laser just kills the weed.â€ Using lasers meant it left no chemical residue in the soil and did not disturb the soil which can lead to carbon being released, he said.
Itâ€™s also more efficient and less costly than labour-intensive alternatives like manual weeding, and minimised crop damage compared to mechanical weeding methods. The

<Files\\Processed x Highly relevant\\2024-08-19 - WKT - Funding sought for futuristic> - § 2 references coded [6.90% Coverage]

Reference 1 - 4.57% Coverage

The Map and Zap system is just what the name suggests: It uses artificial intelligence to recognise weeds and a laser to zap them. In other words, farmers wonâ€™t have to douse their land in weed-killing chemicals.
T

Reference 2 - 2.33% Coverage

He pointed out that while Map and Zap would not replace chemicals, it would reduce the need for them over time.

<Files\\Processed x Highly relevant\\2024-09-11 - MWS - Reducing herbicide usage one z> - § 2 references coded [8.85% Coverage]

Reference 1 - 5.87% Coverage

He has become increasingly concerned with the number of weeds â€" there are at least 40 in New Zealand â€" that have become resistant to herbicides.
In 2016, he started looking into the genetics of weeds to identify how they built up resistance, but a chance meeting with a laser scientist in the United States brought a change of tack.
â€œI asked him if we can take lasers into the field, and he was looking at me like Iâ€™m crazy at that stage ... [he said] to work in agriculture is not impossible, but itâ€™s a big call. Eventually, he came on board.â€

Reference 2 - 2.98% Coverage

â€œIf the big guys of the business come and buy this, then Iâ€™ve done my mission, which is to push the chemical companies to use this as an alternative. â€œTo solve the backyard problems, that takes you off the focus. We should as scientists focus on this, and deliver for farmers.â€

<Files\\Processed x Highly relevant\\2024-09-18 - WHC - Laser focus on killing farm we> - § 2 references coded [8.81% Coverage]

Reference 1 - 1.72% Coverage

APalmerston North man behind a new invention using lasers to kill weeds says it could help stem a reliance on using chemical sprays.
Map an

Reference 2 - 7.09% Coverage

Ghamkhar, director of the Margot Forde Genebank at AgResearch in Palmerston North, said the AI-driven weed identification and laserzapping technology minimised the need for chemical herbicides.
It could be attached to existing farm machinery and used artificial intelligence (AI) to identify unwanted weeds at the early stages of of their lives. It was then zapped by a 300C laser, killing it to the root.
â€œThink of it like â€˜see and sprayâ€™ technology, but using lasers instead of spray. And, without the off-target damage to surrounding crop or pasture,â€ he said.
Map

<Files\\Processed x Highly relevant\\2024-10-01 - GHT - New AI tech aims to aid in sus> - § 4 references coded [20.59% Coverage]

Reference 1 - 4.74% Coverage

Combining artificial intelligence (AI) with traditional knowledge from people who work the land is part of a new AI tool developed to help make decisions that maintain environmental and cultural values. Tupuna Pai is an artificial intelligence-driven platform designed to empower farmers and land stewards or kaitikai of the whenua to make decisions, says founder Phil Kupenga.
Kupenga

Reference 2 - 2.55% Coverage

For example, it can look at data about the soil, water, and weather and then suggest the best times to plant or fertilise crops, or how to reduce the use of harmful chemicals while keeping the farm productive.

Reference 3 - 9.60% Coverage

â€œWhile the final decision will always rest with our farms and land stewards, this platform will help you make choices today that will benefit future generations, all while building a brand that consumers can trust.â€
He said this was trying to solve the problem that many farms and agricultural businesses have â€" making sure their decisions consistently align with their mission and values.
â€œInconsistent approaches to land management not only threaten sustainability but also fail to meet the growing consumer demand for transparency and ethical practices.
â€œTupuna Pai addresses this challenge by integrating Corporate Social Responsibility (CSR) into every decision, ensuring that actions taken on the land are both ethically sound and environmentally responsible. Kupeng

Reference 4 - 3.70% Coverage

â€œIt will minimise environmental and business risks while optimising productivity by being able to instantly consider all risk factors and provide holistic solutions. â€œIt will provide transparency to our community by being able to illustrate the application of tikanga MÄ ori in all things we do.
Â

<Files\\Processed x Highly relevant\\2024-10-14 - NAD - DoC nabs elusive pest with hel> - § 1 reference coded [2.35% Coverage]

Reference 1 - 2.35% Coverage

An elusive Norway rat that was running amok on Motutapu Island in the Hauraki Gulf has been tracked and eliminated with the help of AI technology.

<Files\\Processed x Highly relevant\\2024-11-08 - STF - Keeping an eye on the road as> - § 1 reference coded [2.87% Coverage]

Reference 1 - 2.87% Coverage

The technology coaches drivers to improve their driving skills with instant feedback, to monitoring the condition of a load, and calculates the most optimum routes to ensure greater efficiency and lower carbon dioxide pollution.
â

<Files\\Processed x Highly relevant\\2024-11-13 - PST - NZ firms AI detects disease in> - § 4 references coded [11.37% Coverage]

Reference 1 - 1.81% Coverage

An Auckland deep tech firm is creating artificial intelligence capable of detecting early disease in livestock to help reduce unnecessary death rates.

Reference 2 - 2.54% Coverage

Its technology monitors the health of the herds, and informs farmers and veterinarians for medical intervention. On use in farms, it was found to have reduced the death rate from an average of 4.8% down to 1.4%.

Reference 3 - 2.69% Coverage

â€œWeâ€™re using AI and sensors to basically replicate some of the capabilities of how a human being monitors animals, to provide a 24/7 monitoring solution to farmers for early disease detection,â€ Samiei told The Post.

Reference 4 - 4.33% Coverage

As a byproduct of enabling better health for animals, MACSO wants to work with vets and farmers to reduce the amount of antibiotics consumed by livestock, to produce healthier meat for humans. Separate research by Oxford Analytica showed that controlling disease in livestock could reduce carbon emissions, poverty, hunger and malnutrition, Samiei said.
S

<Files\\Processed x Highly relevant\\2024-12-04 - NZH - The rapid rise and rise of> - § 1 reference coded [2.60% Coverage]

Reference 1 - 2.60% Coverage

Mosgiel-based Techion partnered with AI developers Aware Group to create an AI system that can detect livestock parasites in faecal samples within minutes.
Farmers can act immediately without waiting for results. More importantly, by increasing the accuracy of tests, farmers can reduce unnecessary drenching which is better for livestock health and farm efficiency.